

1. Answer Key: Change Quiz

1. **Answer: (b) 4** — Factor: $\frac{(x-2)(x+2)}{x-2} = x + 2 \rightarrow 4$
2. **Answer: (b) Increasing and concave down** — $f' > 0$ means increasing, $f'' < 0$ means concave down
3. **Answer: (a) 1.5 m/s** — From $x^2 + y^2 = 25$, differentiate: $2x \frac{dx}{dt} + 2y \frac{dy}{dt} = 0$. When $x = 3$, $y = 4$. So $2(3)(2) + 2(4) \frac{dy}{dt} = 0$, giving $\frac{dy}{dt} = -1.5$ m/s (negative = sliding down)
4. **Answer: (b) A local minimum** — $f'(x) = 3x^2 - 6x = 3x(x - 2)$, so critical points at $x = 0, 2$. $f''(x) = 6x - 6$, so $f''(2) = 6 > 0$, meaning $x = 2$ is a local minimum
5. **Answer: (d) Both (b) and (c)** — By chain rule: $f'(x) = \frac{1}{\sin x} \cdot \cos x = \frac{\cos x}{\sin x} = \cot x$
6. **Answer: (a) Check $f(0)$, $f(2)$, and where $f'(x) = 0$** — On a closed interval, extrema occur at critical points OR endpoints
7. **Answer: (a) $\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = 5$** — This is exactly what L'Hôpital's rule states
8. **Answer: (a) $f''(c) = 0$** — When $f''(c) = 0$, the second derivative test is inconclusive; use first derivative test instead
9. **Answer: (c) Must occur at either a critical point or an endpoint** — This is the Extreme Value Theorem in action
10. **Answer: (a) $y = 6x - 9$** — At $x = 3$: $f(3) = 9$, $f'(x) = 2x$, so $f'(3) = 6$. Tangent: $y - 9 = 6(x - 3)$, so $y = 6x - 18 + 9 = 6x - 9$